`RichnessSelection`: math and numerical recipe for optical cluster selection bias

Johnny H. Esteves

Department of Physics, Harvard University

August 5, 2026

The mean lensing surface density around optically selected clusters of observed richness $\lambda^{\rm ob}$ at observed redshift $z^{\rm ob}$ is the sum of a one-halo term (the cluster's own miscentered NFW profile) and a two-halo term (correlated neighbouring halos along the line of sight):

$$\langle\Sigma(R \mid \lambda^{\rm ob}, z^{\rm ob})\rangle \;=\; \langle\Sigma^{\rm 1h}(R \mid \lambda^{\rm ob}, z^{\rm ob})\rangle \;+\; \langle\Sigma^{\rm prj}(R \mid \lambda^{\rm ob}, z^{\rm ob})\rangle, \tag{1}$$

where $\langle\Sigma^{\rm 1h}\rangle$ is evaluated from the miscentered NFW of the target halo and $\langle\Sigma^{\rm prj}\rangle$ is the projected surface density that `RichnessSelection` implements. The physical origin of the selection effect is that redMaPPer counts richness through red-sequence member galaxies, and red galaxies are biased tracers of large-scale structure: their correlation with the surrounding density field means that, at fixed halo mass, a line of sight with more red galaxies gets assigned a larger $\lambda^{\rm ob}$ than one with fewer. The two-halo term at fixed $\lambda^{\rm ob}$ therefore does not carry the bare halo bias $b(M, z)$; Costanzi et al. [2026] show it must instead carry a *selection-affected* cluster bias $b_{\rm sel}(\lambda^{\rm ob}, z^{\rm ob}, \theta)$ that interpolates between a small-scale limit (dominated by close red-galaxy neighbours, $b_{\rm sel}^{\rm small}$) and a large-scale limit (set by correlated LSS at the red-galaxy population bias, $b_{\rm sel}^{\rm large}$). Concretely, rather than writing the two-halo term with the halo-bias aggregate $b_{\rm halo}$ as in a standard lensing pipeline, we factor the scale-dependent selection effect into $b_{\rm sel}$,

$$\langle\Sigma^{\rm prj}(R \mid \lambda^{\rm ob}, z^{\rm ob})\rangle \;=\; \int d\theta\,\sin\theta \int dz\,\frac{dV}{dz\,d\Omega}\int dM\,n(M,z)\big[\,1 + b(M,z)\,b_{\rm sel}(\lambda^{\rm ob}, z^{\rm ob}, \theta)\,\xi_{\rm NL}\,\big]\,\Sigma_{\rm mis}, \tag{2}$$

(the full form with all arguments is eq. (16)). The package returns $b_{\rm sel}$ and the resulting $\langle\Sigma^{\rm prj}(R \mid \lambda^{\rm ob}, z^{\rm ob})\rangle$ at the $(\lambda^{\rm ob}, z^{\rm ob})$ of a DES Y3 cluster bin in ~ 30 ms per call.

This note has three parts. The **model** section gives the mathematical pipeline that assembles $b_{\rm sel}$ non-circularly from halo and projection ingredients, and the resulting two-halo profile $\langle\Sigma^{\rm prj}(R \mid \lambda^{\rm ob}, z^{\rm ob})\rangle$. The **numerical pitfalls** section documents the two subtle hazards we hit when putting that pipeline on a grid — the θ-exclusion step at the halo-halo exclusion boundary and the photo-z band structure of the z-integrand — and the GL-on-physics-split fixes that drive the measured error below 0.01% vs `scipy.integrate.quad`. The **API and glossary** section is a pointer table from each mathematical symbol to the Python call that returns it.

Model

1 Scope and notation

`RichnessSelection` computes the two-halo projected profile $\langle\Sigma^{\rm prj}(R \mid \lambda^{\rm ob}, z^{\rm ob})\rangle$ (eq. 16) together with the selection bias $b_{\rm sel}$ that feeds it, at one $(\lambda^{\rm ob}, z^{\rm ob})$ per call. Photo-z on the target, target miscentering, triaxiality, and REDMAPPER percolation member-loss are left out (only the $\lambda < \lambda^{\rm ob}$ ranking cut is kept). Linear $P(k, z)$ from CAMB feeds $\xi_{\rm NL}$ via halofit + FFTlog.

Masses are in $h^{-1}\,M_\odot$; R is a physical projected distance; θ is a sky angle; χ, R_λ, $R_{\rm excl}$ are comoving in $h^{-1}\,$Mpc. Default cosmology is Buzzard-v1.1 ($\Omega_m = 0.286$, $h = 0.7$, $\sigma_8 = 0.82$). Key shorthand: $R_\lambda(\lambda^{\rm ob}) = (\lambda^{\rm ob}/100)^{0.2}\,h^{-1}\,$Mpc, $R_{\rm excl} \equiv R_\lambda(\lambda^{\rm ob})(1 + z^{\rm ob})$, and $b_{\rm sel}^{\rm small}/b_{\rm sel}^{\rm large}$ are the two asymptotes of the sigmoid ansatz below.

2 Core equations

2.1 Halo-bias aggregate $b_{\rm halo}(\lambda^{\rm ob}, z^{\rm ob})$

A single 1-D $\ln M$ integral at fixed $z^{\rm ob}$, weighting the Tinker-2010 halo bias by the halo-mass distribution at fixed observed richness:

$$\boxed{b_{\rm halo}(\lambda^{\rm ob}, z^{\rm ob}) = \frac{\int d\ln M\ \ M\, n(M, z^{\rm ob})\, b(M, z^{\rm ob})\, P(\lambda^{\rm ob} \mid M, z^{\rm ob})}{\int d\ln M\ \ M\, n(M, z^{\rm ob})\, P(\lambda^{\rm ob} \mid M, z^{\rm ob})}.} \tag{3}$$

Here M is the halo mass in $h^{-1}\,M_\odot$; $\ln M$ runs over the integration support $[10^{13}, 10^{15.5}]\,h^{-1}\,M_\odot$; $n(M, z)$ is the Tinker-2008 halo mass function (§6); $b(M, z)$ is the Tinker-2010 peak-height halo bias; and $P(\lambda^{\rm ob} \mid M, z)$ is the mass–observable relation pdf from `MOR`. The factor of M inside the integrand converts dM to $d\ln M$.

2.2 Sigmoid ansatz for $b_{\rm sel}(\theta)$

Costanzi et al. [2026], eq. `b_sel_theta`:

$$\boxed{b_{\rm sel}(\lambda^{\rm ob}, \lambda^{\rm tr}, z^{\rm ob}, \theta) = b_{\rm sel}^{\rm small}\,[\,1 - \sigma(\theta)\,] + b_{\rm sel}^{\rm large}\,\sigma(\theta), \qquad \sigma(\theta) = \frac{1}{1 + e^{-k(\theta-\theta_0)}}.} \tag{4}$$

Here θ is the sky angular separation from the target; $b_{\rm sel}^{\rm small}$ and $b_{\rm sel}^{\rm large}$ are the two plateaus (small- and large-angle limits of the bias, both scalars at fixed $(\lambda^{\rm ob}, \lambda^{\rm tr}, z^{\rm ob})$); $\sigma(\theta)$ is the logistic transition function whose shape constants are fixed to $k = 2.5/\theta_\lambda$ and $\theta_0 = \theta_\lambda/2$, with $\theta_\lambda \equiv R_\lambda(\lambda^{\rm ob})/D_A(z^{\rm ob})$ the angle subtended by the target's REDMAPPER radius $R_\lambda(\lambda^{\rm ob}) = (\lambda^{\rm ob}/100)^{0.2}\,h^{-1}\,$Mpc. The θ-shape is fixed analytically; only the two plateaus $b_{\rm sel}^{\rm small}$ and $b_{\rm sel}^{\rm large}$ need to be computed.

2.3 Projection kernel and $\mathcal{P}[X]$ operator

The *projection kernel* $\rho^{\rm prj}$ is the per-halo weight with which a projected halo at (λ, z, θ) contaminates the observed richness of a target at $(\lambda^{\rm ob}, z^{\rm ob})$ (Costanzi et al. 2026, eq. `weight`):

$$\rho^{\rm prj}(\lambda, z, \theta \mid \lambda^{\rm ob}, z^{\rm ob}) \;=\; w_z(z, z^{\rm ob})\, f_A(\theta, \lambda^{\rm ob}, z^{\rm ob}, \lambda, z)\, \lambda, \qquad 0 < \lambda < \lambda^{\rm ob}, \tag{5}$$

and $\rho^{\rm prj} \equiv 0$ for $\lambda \geq \lambda^{\rm ob}$ — the REDMAPPER ranking cut: only projected halos of *lower* richness than the target contaminate the target's aperture. Rather than carry an indicator function through

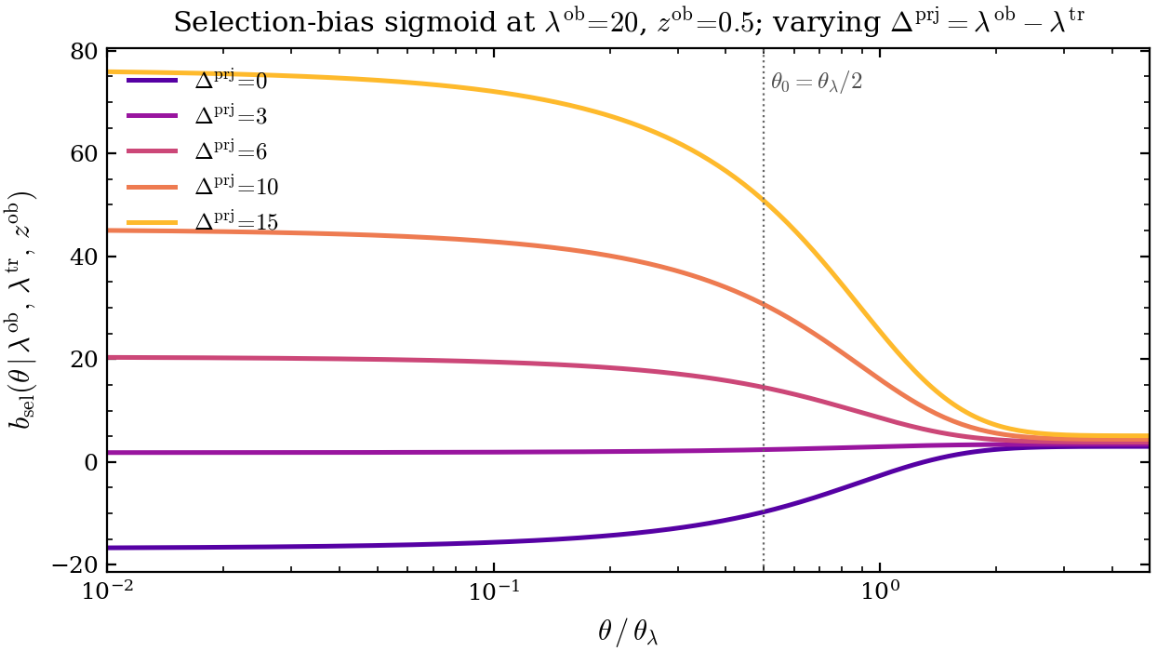

Figure 1: Selection-bias sigmoid $b_{\rm sel}(\theta\,|\,\lambda^{\rm ob},\lambda^{\rm tr},z^{\rm ob})$ at fixed $(\lambda^{\rm ob},z^{\rm ob}) = (20,0.5)$ as we vary the projection contamination $\Delta^{\rm prj} \equiv \lambda^{\rm ob} - \lambda^{\rm tr}$. The x-axis is scaled by the per-bin angle θ_λ, so the sigmoid pivot $\theta_0 = \theta_\lambda/2$ (dotted) is universal. The θ-shape is fixed by the ansatz; $\Delta^{\rm prj}$ drives both plateaus through Steps 4–5. The left plateau $b_{\rm sel}^{\rm small}$ (close neighbours projecting into the target aperture) grows fastest with $\Delta^{\rm prj}$; the right plateau $b_{\rm sel}^{\rm large}$ (correlated large-scale structure) moves less because it is closed by the empirical $b_{\rm halo}[1 + 0.13\,\delta^{\rm prj}]$ relation.

every equation below, this cut is stated once, here, and from now on enforced silently by writing every λ-integral with upper limit $\lambda^{\rm ob}$ (never ∞). Here λ is the projected halo's true richness; z is its redshift; θ is its angular separation from the target; $w_z(z,z^{\rm ob})$ is the parabolic photo-z kernel quantifying the probability that a galaxy at redshift z is counted in the target's aperture at $z^{\rm ob}$; $f_A(\theta,\lambda^{\rm ob},z^{\rm ob},\lambda,z) \in [0,1]$ is the closed-form overlap fraction of the two redMaPPer disks (of the target and the projected halo); and the factor λ converts "this halo is present" into "this halo contributes λ counts".

Costanzi et al. 2026 (`papers/Costanzi2026.tex`, § "Projection kernel", Fig. 1) states the same weight per projected halo i as $\rho_i^{\rm prj} = \lambda_i\, w(z_i,z^{\rm ob})\, f_{A,i}$: both $w(z_i,z^{\rm ob})$ — the photo-z kernel above, which sets how much line-of-sight weight the far zone carries and is *not* negligible — and the geometric factor $f_{A,i}$ are needed; this subsection derives the latter.

The overlap fraction f_A encodes the redMaPPer membership geometry: the model treats the projected halo's λ member galaxies as uniformly distributed over the halo's own richness disk, of angular radius $\theta_\lambda(\lambda,z) = R_\lambda(\lambda)(1+z)/\chi(z)$; the target gains only those galaxies that fall inside *its* disk, of radius $\theta_{\lambda,\rm ob} \equiv \theta_\lambda(\lambda^{\rm ob},z^{\rm ob})$. For a centre separation θ the expected gained count is therefore λ times the fraction of the projector's disk area that overlaps the target's:

$$f_A(\theta,\lambda^{\rm ob},z^{\rm ob},\lambda,z) \;=\; \frac{A_{\rm ov}(\theta,\theta_{\lambda,\rm ob},\theta_\lambda(\lambda,z))}{\pi\,\theta_\lambda^2(\lambda,z)}, \tag{6}$$

with $A_{\rm ov}$ the circle–circle intersection ("lens") area of the target and projector disks at centre distance

θ,

$$A_{\rm ov}(\theta,\theta_{\lambda,\rm ob},\theta_\lambda) = \theta_{\lambda,\rm ob}^2 \arccos\Big(\frac{\theta^2+\theta_{\lambda,\rm ob}^2-\theta_\lambda^2}{2\theta\,\theta_{\lambda,\rm ob}}\Big) + \theta_\lambda^2 \arccos\Big(\frac{\theta^2+\theta_\lambda^2-\theta_{\lambda,\rm ob}^2}{2\theta\,\theta_\lambda}\Big) - \tfrac{1}{2}\sqrt{\big[(\theta_{\lambda,\rm ob}+\theta_\lambda)^2-\theta^2\big]\big[\theta^2-(\theta_{\lambda,\rm ob}-\theta_\lambda)^2\big]} \tag{7}$$

for $|\theta_{\lambda,\rm ob}-\theta_\lambda| < \theta < \theta_{\lambda,\rm ob}+\theta_\lambda$, with $A_{\rm ov} = \pi\min(\theta_{\lambda,\rm ob},\theta_\lambda)^2$ for $\theta \le |\theta_{\lambda,\rm ob}-\theta_\lambda|$ (full containment) and $A_{\rm ov} = 0$ for $\theta \ge \theta_{\lambda,\rm ob}+\theta_\lambda$ (no contact). Note the normalisation: the denominator of eq. (6) is the *projector's* own disk area, not the target's, because f_A is the fraction of the projector's galaxies captured by the target aperture. This makes $f_A \in [0,1]$ exact, with $f_A = 1$ when the projector disk sits entirely inside the target's ($\theta \le \theta_{\lambda,\rm ob}-\theta_\lambda$, always reachable at $z \simeq z^{\rm ob}$ since $\lambda < \lambda^{\rm ob}$ implies $\theta_\lambda < \theta_{\lambda,\rm ob}$) and $f_A = 0$ beyond $\theta_{\lambda,\rm ob}+\theta_\lambda$ — so the θ-integral support is finite. $A_{\rm ov}$ is homogeneous of degree two in its three angular arguments, hence f_A depends only on the ratio $\theta_\lambda/\theta_{\lambda,\rm ob}$ — the property the frozen-physics companion note (`docs/richness_selection_frozen.tex`, §4) exploits. Implementation: `richness_selection.geometry.area_overlap`, which always normalises by the *second* disk's area regardless of which disk is larger; the appendix-A formula of Costanzi et al. 2026 instead normalises by $\pi\min(\theta_{\lambda,\rm ob},\theta_\lambda)^2$, so the two disagree when a low-z foreground halo's own disk exceeds the target's ($\theta_\lambda > \theta_{\lambda,\rm ob}$, possible even for $\lambda < \lambda^{\rm ob}$ since angular size grows as $z \to 0$).

The $\mathcal{P}[X]$ *operator* is the line-of-sight-averaged integral of an arbitrary $X(M,z,\theta)$ weighted by the halo distribution and $\rho^{\rm prj}$:

$$\mathcal{P}[X](\lambda^{\rm ob},z^{\rm ob}) \;\equiv\; \int dz\,\frac{dV}{dz\,d\Omega}(z)\int dM\,n(M,z)\int d\lambda\,P(\lambda\mid M,z) \cdot\; 2\pi\int d\theta\,\sin\theta\;\rho^{\rm prj}(\lambda,z,\theta\mid\lambda^{\rm ob},z^{\rm ob})\,X(M,z,\theta\mid z^{\rm ob}). \tag{8}$$

The integrations marginalise, in order, over the projected halo's redshift z (weighted by comoving volume per steradian $dV/dz\,d\Omega$), mass M (weighted by the halo mass function $n(M,z)$), true richness λ (weighted by the MOR $P(\lambda\mid M,z)$), and angular separation θ from the target (with the usual $2\pi\sin\theta\,d\theta$ sky-element).

Three specialisations drive the pipeline:

$$\mathcal{P}[1] \;=\; \langle\Delta^{\rm prj}_{\rm bkg}\rangle(\lambda^{\rm ob},z^{\rm ob}), \qquad I_2 \;\equiv\; \mathcal{P}[b\,\xi_{\rm NL}]\,, \qquad I_1 \;\equiv\; \mathcal{P}[b\,\xi_{\rm NL}\,\sigma(\theta)]\,, \tag{9}$$

where $b = b(M,z)$ is the Tinker-2010 halo bias (the halo's own clustering amplitude) and $\xi_{\rm NL} = \xi_{\rm NL}(z,z^{\rm ob},\theta)$ is the nonlinear matter correlation function evaluated at the 3-D comoving separation $\Delta\chi(z,z^{\rm ob},\theta)$ between the target at $z^{\rm ob}$ and the projected halo at (z,θ). By linearity, $\mathcal{P}[b\,\xi_{\rm NL}(1-\sigma)] = I_2 - I_1$. Halo exclusion sets $\xi_{\rm NL}\equiv 0$ whenever $\Delta\chi < R_{\rm excl}$.

2.4 Bias pipeline

This subsection gives the calculation chain from halo and projection ingredients to the selection-affected cluster bias $b_{\rm sel}(\lambda^{\rm ob},z^{\rm ob},\theta)$ that eq. (16) requires. From the halo ingredients $\{n(M,z), b(M,z), P(\lambda\mid M,z)\}$ and the projection ingredients $\{w_z, f_A, \xi_{\rm NL}\}$ it first evaluates the halo-bias aggregate $b_{\rm halo}(\lambda^{\rm ob},z^{\rm ob})$ of eq. (3), a 1-D $\ln M$ integral at fixed $z^{\rm ob}$. In parallel, the hardest numerical block computes the three forward integrals $\mathcal{P}[1]$, I_1, I_2 of eq. (8); the recipes for this step are the subject of §3–4 below. With these in hand, the random-aperture projection contamination

$$\Delta^{\rm prj}_{\rm RND}(\lambda^{\rm ob},z^{\rm ob}) \;=\; \mathcal{P}[1](\lambda^{\rm ob},z^{\rm ob}) + b_{\rm halo}(\lambda^{\rm ob},z^{\rm ob})\,I_2(\lambda^{\rm ob},z^{\rm ob}) \tag{10}$$

follows by substituting $b_{\rm sel} \to b_{\rm halo}$ in the $b_{\rm sel}$-linearised identity ($b_{\rm halo}$ pulls out of the mass integral because it is mass-independent). The large-θ plateau is then

$$b_{\rm sel}^{\rm large}(\lambda^{\rm ob}, \lambda^{\rm tr}, z^{\rm ob}) \;=\; b_{\rm halo}(\lambda^{\rm ob}, z^{\rm ob}) \left[\, 1 + 0.13\, \delta^{\rm prj}(\lambda^{\rm ob}, \lambda^{\rm tr}, z^{\rm ob}) \,\right], \qquad \delta^{\rm prj} \;\equiv\; \frac{\lambda^{\rm ob} - \lambda^{\rm tr}}{\Delta^{\rm prj}_{\rm RND}} - 1, \tag{11}$$

where $\delta^{\rm prj}$ is the fractional excess contamination at $(\lambda^{\rm ob}, \lambda^{\rm tr}, z^{\rm ob})$ relative to a random aperture and the empirical coefficient 0.13 is the Buzzard calibration. Finally the small-θ plateau closes the remaining linear equation in $b_{\rm sel}$:

$$b_{\rm sel}^{\rm small}(\lambda^{\rm ob}, \lambda^{\rm tr}, z^{\rm ob}) \;=\; \frac{(\lambda^{\rm ob} - \lambda^{\rm tr}) - \mathcal{P}[1] - b_{\rm sel}^{\rm large}\, I_1}{I_2 - I_1}, \tag{12}$$

where the denominator $I_2 - I_1 = \mathcal{P}[b\, \xi_{\rm NL}(1-\sigma)]$ is the contribution of the $1-\sigma$ (small-θ) part of the sigmoid ansatz, and the numerator is what remains of the $\Delta^{\rm prj}$ identity once the $\mathcal{P}[1]$ constant and the $b_{\rm sel}^{\rm large}$-weighted I_1 piece have been subtracted. With $b_{\rm sel}^{\rm small}$ and $b_{\rm sel}^{\rm large}$ both known, eq. (4) returns $b_{\rm sel}(\lambda^{\rm ob}, \lambda^{\rm tr}, z^{\rm ob}, \theta)$. Marginalising over $\lambda^{\rm tr}$ via

$$b_{\rm sel}(\lambda^{\rm ob}, z^{\rm ob}, \theta) \;=\; \int d\lambda^{\rm tr}\, b_{\rm sel}(\lambda^{\rm ob}, \lambda^{\rm tr}, z^{\rm ob}, \theta)\, P(\lambda^{\rm tr} \mid \lambda^{\rm ob}, z^{\rm ob}), \tag{13}$$

with $P(\lambda^{\rm tr} \mid \lambda^{\rm ob}, z^{\rm ob}) \propto P(\lambda^{\rm ob} \mid \lambda^{\rm tr}, z^{\rm ob})\, \pi(\lambda^{\rm tr}, z^{\rm ob})$ and $\pi(\lambda^{\rm tr}, z^{\rm ob}) = \int dM\, n(M,z)\, P(\lambda^{\rm tr} \mid M, z)$ the prior on the true richness, gives the final input to eq. (16).

Because eq. (4) is linear in the plateaus $(b_{\rm sel}^{\rm small}, b_{\rm sel}^{\rm large})$ and the sigmoid $\sigma(\theta)$ carries no $\lambda^{\rm tr}$ dependence, the $\lambda^{\rm tr}$-marginalisation commutes with the sigmoid assembly. Defining the $\lambda^{\rm tr}$-averaged plateaus

$$\bar{b}_{\rm small}(\lambda^{\rm ob}, z^{\rm ob}) \;=\; \int d\lambda^{\rm tr}\, b_{\rm sel}^{\rm small}(\lambda^{\rm ob}, \lambda^{\rm tr}, z^{\rm ob})\, P(\lambda^{\rm tr} \mid \lambda^{\rm ob}, z^{\rm ob}), \qquad \bar{b}_{\rm large}(\lambda^{\rm ob}, z^{\rm ob}) \;=\; \int d\lambda^{\rm tr}\, b_{\rm sel}^{\rm large}(\lambda^{\rm ob}, \lambda^{\rm tr}, z^{\rm ob})\, P(\lambda \tag{14}$$

eq. (13) reduces to

$$\boxed{b_{\rm sel}(\lambda^{\rm ob}, z^{\rm ob}, \theta) \;=\; \bar{b}_{\rm small}(\lambda^{\rm ob}, z^{\rm ob})\, [\, 1 - \sigma(\theta)\,] + \bar{b}_{\rm large}(\lambda^{\rm ob}, z^{\rm ob})\, \sigma(\theta).} \tag{15}$$

Eq. (15) is numerically identical to eq. (13) (it is an algebraic rearrangement, not an approximation), but shifts the $\lambda^{\rm tr}$-quadrature from *inside* the θ-loop to a one-shot pre-computation at fixed $(\lambda^{\rm ob}, z^{\rm ob})$. Downstream consumers that assemble $b_{\rm sel}(\theta)$ at many θ-nodes — notably $\langle \Sigma^{\rm prj} \rangle$ and $\langle \Delta\Sigma^{\rm prj} \rangle$ — therefore only pay the cost of the analytic sigmoid per call.

The $b_{\rm sel}^{\rm small}$-denominator guard in eq. (12) ($|I_2 - I_1| \to 0 \Rightarrow b_{\rm sel}^{\rm small} \leftarrow b_{\rm sel}^{\rm large}$) is applied per-$\lambda^{\rm tr}$ before the average, so the definition of $\bar{b}_{\rm small}$ in eq. (14) is well-defined even when the denominator is small for a subset of the $\lambda^{\rm tr}$ grid.

In `SelBias` (§4), steps computed once per $(\lambda^{\rm ob}, z^{\rm ob})$ ($b_{\rm halo}$, $\mathcal{P}[1]$, I_1, I_2, $\Delta^{\rm prj}_{\rm RND}$) are cached inside `bias_precompute`; the per-$\lambda^{\rm tr}$ closures of eqs. (11)–(12) live in `bias_from_precomp`; the sigmoid assembly at fixed $\lambda^{\rm tr}$ is `b_lob_theta`; and the $\lambda^{\rm tr}$-marginalised plateaus $(\bar{b}_{\rm small}, \bar{b}_{\rm large})$ of eq. (14) are cached by `_marginalised_plateaus`. The public entry point `b_sel_marginalised(theta, lob, zob)` evaluates eq. (15) directly from those cached plateaus.

2.5 Two-halo projected profile

Costanzi et al. [2026], eq. `sigprj`:

$$\langle \Sigma^{\rm prj}(R \mid \lambda^{\rm ob}, z^{\rm ob}) \rangle = \int d\theta\, \sin\theta \int dz\, \frac{dV}{dz\, d\Omega} \int dM\, n(M,z) \cdot \left[\, 1 + b(M,z)\, b_{\rm sel}(\lambda^{\rm ob}, z^{\rm ob}, \theta)\, \xi_{\rm NL}(z, z^{\rm ob}, \theta) \right] \Sigma_{\rm mis}(R \mid M, z, \theta, z^{\rm ob}). \tag{16}$$

Here R is the physical projected distance around the target in $h^{-1}\,$Mpc; θ is the angular separation of the contaminating halo from the target; z is the halo's redshift; M its mass; $n(M,z)$ the halo mass function; $b(M,z)$ the halo bias; $b_{\rm sel}$ the selection-affected cluster bias from eq. (13); $\xi_{\rm NL}$ the nonlinear matter correlation function at the 3-D separation between the target and the (z,θ) halo; and $\Sigma_{\rm mis}$ is the azimuth-averaged miscentered NFW surface density. The 1 inside the bracket is the mean (uncorrelated) contribution; the $b\,b_{\rm sel}\,\xi_{\rm NL}$ term is the correlation-function boost that the two-halo term captures.

$\Sigma_{\rm mis}$ is the one-halo surface density integrated over the azimuthal angle φ because the projected halo is off-centre with respect to the profile radius:

$$\Sigma_{\rm mis}(R \mid M, z, \theta, z^{\rm ob}) = \int_0^{2\pi} d\varphi\; \Sigma(R_h \mid M, z), \qquad R_h = \sqrt{R^2 + R_\theta^2 - 2RR_\theta\cos\varphi}, \tag{17}$$

with $R_\theta \equiv \theta\, D_A(z^{\rm ob})$ the physical projected distance from the target to the contaminating halo and $\Sigma(R_h \mid M, z)$ the centred NFW surface density of the contaminating halo at 2-D radius R_h from its own centre. $\Sigma_{\rm mis}$ is tabulated from the offset-NFW lookup (`nfw.py`), with a factor of 2 to match the $\int_0^{2\pi}$ convention.

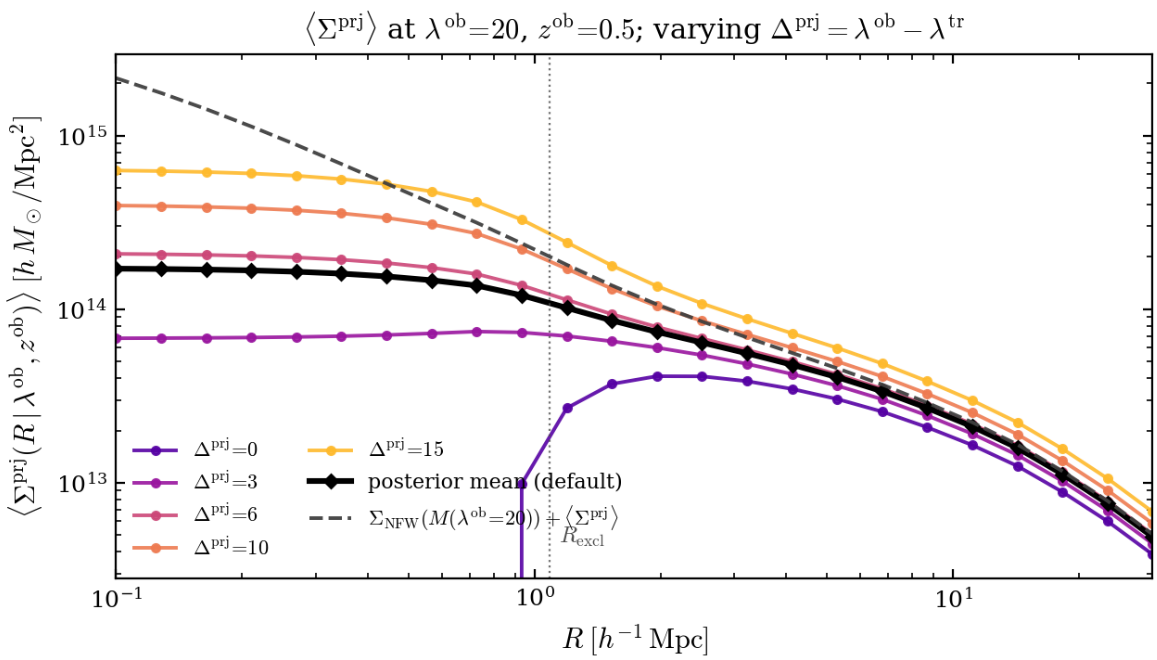

Figure 2: Two-halo $\langle\Sigma^{\rm prj}(R \mid \lambda^{\rm ob}, \lambda^{\rm tr}, z^{\rm ob})\rangle$ at $(\lambda^{\rm ob}, z^{\rm ob}) = (20, 0.5)$, varying the projection contamination $\Delta^{\rm prj} \equiv \lambda^{\rm ob} - \lambda^{\rm tr}$ (same colour palette as Fig. 1). $b_{\rm sel}$ is evaluated at fixed $\lambda^{\rm tr}$ (not $\lambda^{\rm tr}$-marginalised) to match the Fig. 1 exercise. Dotted line: $R_{\rm excl} = R_\lambda(\lambda^{\rm ob})(1+z^{\rm ob})$, separating the miscentered-NFW-dominated inner regime from the correlated $b_{\rm sel}{\cdot}\xi_{\rm NL}$ large-scale regime. $\Delta^{\rm prj} = 0$ sits anomalously low at small R because both $b_{\rm sel}$ plateaus collapse to the halo-bias aggregate in that limit.

2.6 Lensing excess $\langle\Delta\Sigma^{\rm prj}(R \mid \lambda^{\rm ob}, z^{\rm ob})\rangle$

The two-halo observable returned by `DeltaSigmaPrj` is the radial excess of (16),

$$\langle\Delta\Sigma^{\rm prj}(R \mid \lambda^{\rm ob}, z^{\rm ob})\rangle \;\equiv\; \bar\Sigma^{\rm prj}(<R \mid \lambda^{\rm ob}, z^{\rm ob}) \;-\; \langle\Sigma^{\rm prj}(R \mid \lambda^{\rm ob}, z^{\rm ob})\rangle. \tag{18}$$

Because $R \to \bar\Sigma(<R) - \Sigma(R)$ is a linear functional in the radial argument only, it commutes with the (θ, z, M) integrals in eq. (16): the pipeline of §3–4 carries over unchanged, and the only substi-

tution is at the NFW lookup, $\Sigma_{\rm mis} \to \Delta\Sigma_{\rm mis} \equiv \bar{\Sigma}_{\rm mis}(<R) - \Sigma_{\rm mis}(R)$, evaluated via the companion `log_deltasigma_single` table shipped with the offset-NFW kit. The full derivation, the scale argument for adaptive $\theta_{\max}$, and the factor-of-2 convention cross-check against the analytical Wright & Brainerd 2000 centred NFW $\Delta\Sigma$ live in `docs/delta_sigma_prj_derivation.tex`.

Profile vs projection contamination. Figure 3 shows $\langle \Delta\Sigma^{\rm prj}(R \mid \lambda^{\rm ob}, \lambda^{\rm tr}, z^{\rm ob})\rangle$ at $(\lambda^{\rm ob}, z^{\rm ob}) = (20, 0.5)$, varying $\Delta^{\rm prj} = \lambda^{\rm ob} - \lambda^{\rm tr}$ — the same sweep as Fig. 1 ($b_{\rm sel}$-sigmoid) and Fig. 2 ($\langle\Sigma^{\rm prj}\rangle$). The shape is the familiar two-halo lensing bump peaking around $\sim R_\lambda$ (dotted $R_{\rm excl}$ marker); amplitude scales with $\Delta^{\rm prj}$ because both $b_{\rm sel}$ plateaus ($b_{\rm sel}^{\rm small}, b_{\rm sel}^{\rm large}$) grow with projection contamination through Steps 4–5 of the bias pipeline.

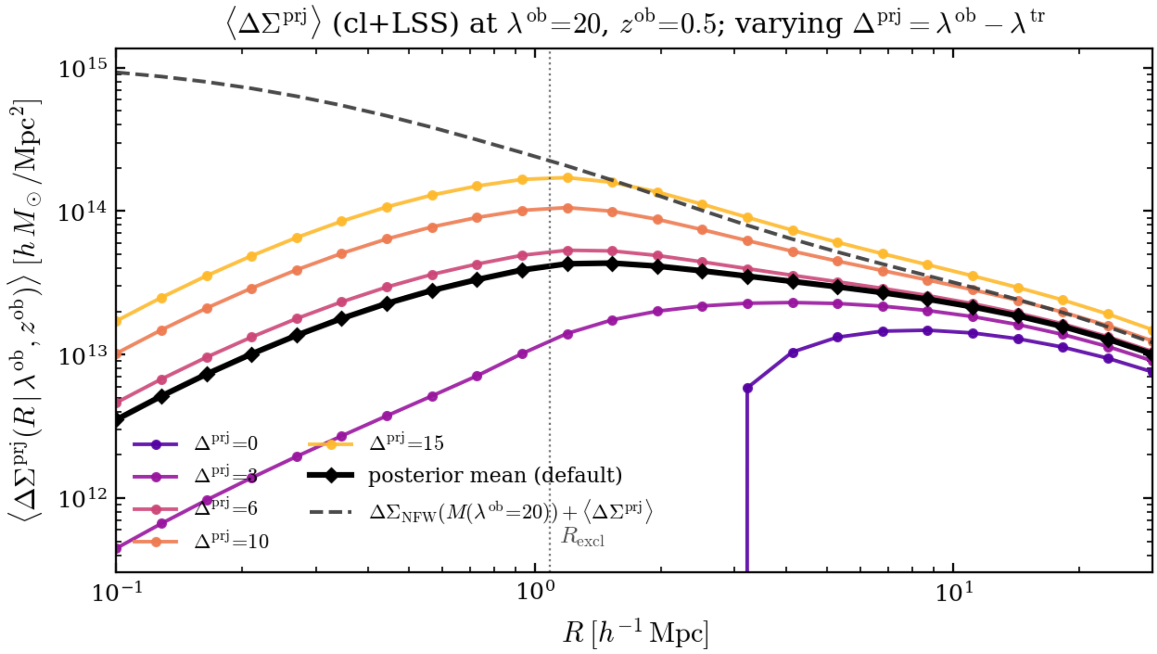

Figure 3: $\langle\Delta\Sigma^{\rm prj}(R\,|\,\lambda^{\rm ob}, \lambda^{\rm tr}, z^{\rm ob})\rangle$ at $(\lambda^{\rm ob}, z^{\rm ob}) = (20, 0.5)$, varying $\Delta^{\rm prj} = \lambda^{\rm ob} - \lambda^{\rm tr}$ (`cl+LSS` correlation-excess piece, the class default). Dotted line: $R_{\rm excl} = R_\lambda(\lambda^{\rm ob})(1 + z^{\rm ob})$. Inner cut-off from the LoS-slab exclusion; outer decay set by the joint support of $\Delta\Sigma_{\rm mis}(R \mid R_{\rm mis}\!\sim\! R)$ and $\xi_{\rm NL}(r)$.

Precision vs `scipy.quad`. Figure 4 compares the production `DeltaSigmaPrj` ($n_{\rm per\_seg} = 30$, $N_M = 24$, $N_z = 80$, $R_{\max} = 30\,h^{-1}\,{\rm Mpc}$) against a nested-`scipy.quad` reference built by `validations/delta_sigma_prj_diagnostics.py` at the same $(\lambda^{\rm ob}, z^{\rm ob}) = (20, 0.5)$ and $R \in \{0.3, 1, 3, 10\}\,h^{-1}\,{\rm Mpc}$. The reference is parallelised across the 8 $(R, {\rm kind})$ jobs via `multiprocessing.Pool` (8-core wall clock ~ 3 min vs ~ 35 min serial); quad tolerances $\epsilon_\theta = 3\times10^{-3}$, $\epsilon_z = 10^{-2}$. The residual panel shows sub-0.5% agreement on $R \geq 1\,h^{-1}\,{\rm Mpc}$ on both `total` and `cl`. At $R = 0.3$ the residual reaches $\sim 2\%$: the $[1 + b\,b_{\rm sel}\,\xi_{\rm NL}]$ bracket changes amplitude sharply where $\theta_{\rm excl}(z)$ and $\theta_R = R/D_A(z^{\rm ob})$ coincide, and `scipy.quad`'s adaptive refinement issues an `IntegrationWarning` at that crossing. Tightening the quad tolerances by 3× shifts the reference by $< 0.01\%$ (so the quad itself is converged), and the GL-on-segments production side is self-converged to $< 0.5\%$ between $n_{\rm per\_seg} = 30$ and 120: the residual at $R = 0.3$ is therefore a real quadrature-method difference at the smallest R, not a bug in either side. The regression test (`tests/test_delta_sigma_prj.py`) encodes this as a per-R tolerance (0.5% on $R \geq 1$, 2.5% at $R = 0.3$).

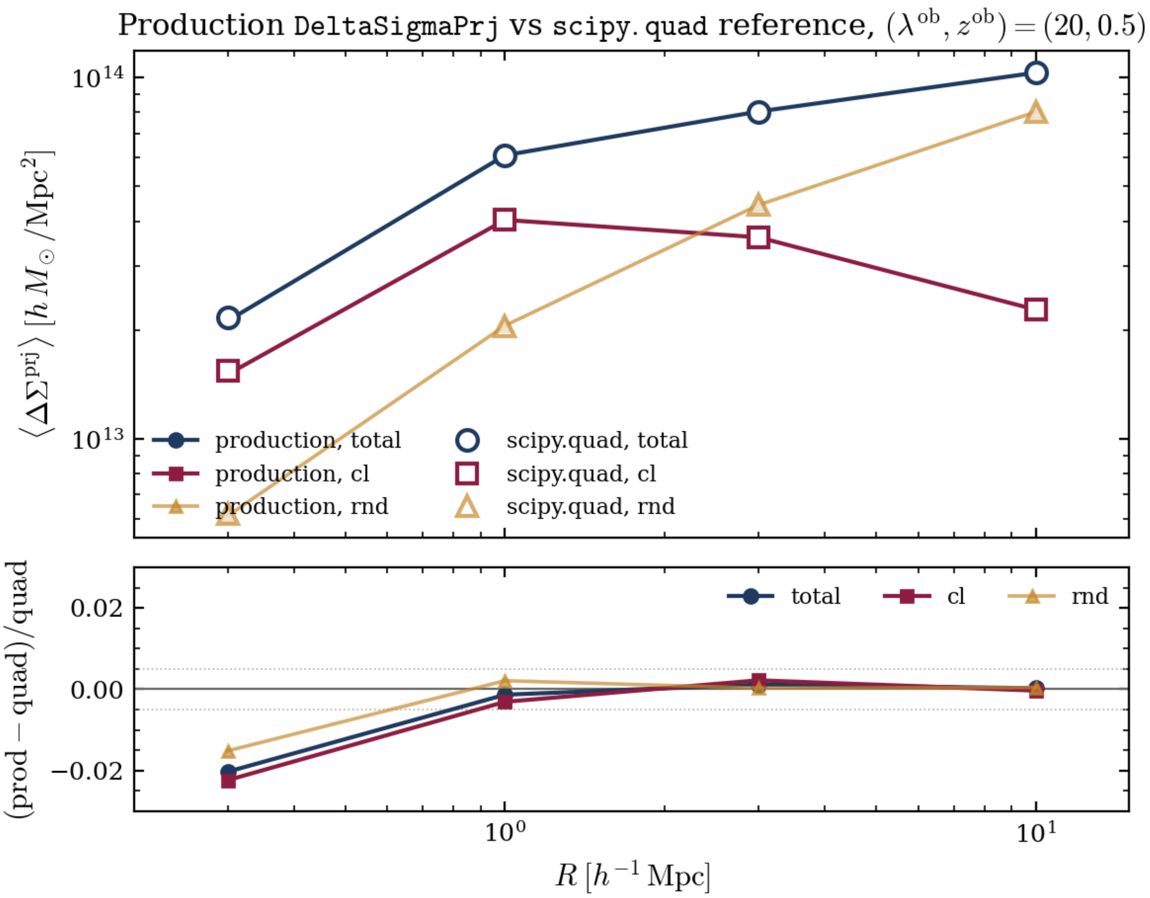

Figure 4: Precision audit of production `DeltaSigmaPrj` (filled markers, solid) against the `scipy.quad` reference (open markers) at $(\lambda^{\rm ob}, z^{\rm ob}) = (20, 0.5)$. Top: absolute profiles for `total`, `cl` (default return), and `rnd` (truncation-dependent boundary term). Bottom: fractional residual (prod−quad)/quad, with $\pm 0.5\%$ guides (dotted). Agreement is sub-0.5% for $R \geq 1\,h^{-1}$ Mpc; at $R = 0.3$ the $\sim 2\%$ residual tracks a real quadrature-method difference where $\theta_{\rm excl}(z) \sim \theta_R$ (see text).

Numerical pitfalls

The three forward integrals that carry the whole pipeline — $\mathcal{P}[1]$, I_1, I_2 of eq. (8) — are 4-D integrals over (z, θ, M, λ). The inner (M, λ) axes are well-behaved and a plain Gauss–Legendre (GL) rule with $N_M = 24$ and $N_\lambda = 60$ converges them to $< 10^{-4}$; that part of the integration is boring. The outer (z, θ) axes, however, contain two sharp, physics-driven features that punish any uninformed fixed-rule method and were the source of the largest precision bugs we saw while developing this package:

- the **halo-exclusion step in** θ: at each z the integrand has a hard discontinuity at $\theta = \theta_{\rm excl}(z)$, where $\xi_{\rm NL}$ is set to zero by the exclusion prescription. A fixed GL grid on $(0, 2\theta_\lambda)$ with a mask integrates across this step and stalls at $\sim 0.3\%$ error regardless of N_θ (§3);
- the **exclusion-peak + ring-plateau structure in** z: the photo-z kernel carves out a window of width ~ 0.1 around $z^{\rm ob}$, and inside that window $\xi_{\rm NL}$ produces twin sharp peaks at $z = z^{\rm ob} \pm \delta z_{\rm excl}$ sitting on top of a narrow plateau — three qualitatively different regions crammed into $|z - z^{\rm ob}| \lesssim 5 \times 10^{-4}$ (§4).

The recipes below move the GL nodes onto the physically correct coordinate in each region so that *the sharp features land at grid boundaries*, where GL rules naturally cluster nodes. The result is sub-0.01% precision on all three integrals at $N_z = 80$, $N_\theta = 10$ vs `scipy.integrate.quad` with matched inner grids. Total cost at $(\lambda^{\rm ob}, z^{\rm ob}) = (20, 0.5)$: $\sim$ 26 ms per `SelBias.bias_precompute`. If the bullets below already look familiar, the executive-summary version is: split the θ-axis at the exclusion boundary (per-z GL on $[\theta_{\rm excl}(z),\, 2\theta_\lambda]$, no mask); split the z-axis by physics into a ring band + two outer halves, integrating in z on the ring and in $u = \ln|\Delta\chi_\parallel|$ on the outers.

3 The θ-axis: split at exclusion

What this section fixes. The θ-axis of $\mathcal{P}[1], I_1, I_2$ runs over the angular separation between the target and each projected halo. Halo exclusion makes this a discontinuous integrand, which breaks any fixed GL rule that sees the discontinuity as a noisy feature. Below we show why that happens and how moving the lower integration limit onto the discontinuity turns the problem into a smooth integral whose GL convergence is at its design rate.

At any z, halo exclusion kills θ below a threshold $\theta_{\rm excl}(z)$ obtained from $\Delta\chi(z, \theta_{\rm excl}) = R_{\rm excl}$:

$$\cos\theta_{\rm excl}(z) = \frac{\chi(z)^2 + \chi(z^{\rm ob})^2 - R_{\rm excl}^2}{2\,\chi(z)\,\chi(z^{\rm ob})}, \qquad \text{clipped to } [-1, 1]. \tag{19}$$

A fixed-grid $N_\theta = 20$ GL on $(0, 2\theta_\lambda)$ with $\xi_{\rm NL} \to 0$ inside the step integrates across a discontinuity: the error oscillates in N_θ and settles to a $\sim$ 0.3% floor. The fix is a per-z change of the integration interval: when $\theta_{\rm excl}(z) > 0$, integrate on $[\theta_{\rm excl}(z),\, 2\theta_\lambda]$ with no mask. GL nodes cluster at the endpoints where the step sits; the integrand is smooth on the interval.

N_θ	$\mathcal{P}[1]$ err	I_2 err	I_1 err
5	+0.060%	+0.046%	+0.046%
10	**−0.005%**	**−0.001%**	**−0.004%**
15	+0.000%	+0.000%	+0.000%

$N_\theta = 10$ reaches sub-0.01%: $\sim 12\times$ node reduction and two decades better precision than the mask recipe.

4 The z-axis: split by physics

What this section fixes. With θ handled, the outer z-integral becomes the dominant source of residual error — not because it is hard in absolute terms, but because its integrand has three different shapes over a $\sim 10^{-3}$-wide window around $z^{\rm ob}$. A single uniform z-grid under-samples the exclusion peaks; a single log-spaced grid in $|\Delta\chi_\parallel|$ over-concentrates on the inner edge and misses the ring plateau. The fix is to integrate each physical region on the coordinate in which *that* region is smooth, and stitch them together.

Each of $\mathcal{P}[1], I_1, I_2$ factorises into an outer z-integral times an *inner* (M, λ, θ)-integral,

$$I \;=\; \int dz\, f_X(z), \qquad f_X(z) \;\equiv\; \frac{dV}{dz\, d\Omega}(z)\, w_z(z, z^{\rm ob})\, \mathcal{I}^X_{\rm inner}(z), \tag{20}$$

where the inner integral is the full (M, λ, θ) block of eq. (8) evaluated at a single z,

$$\mathcal{I}^X_{\rm inner}(z) \;\equiv\; \int dM\, n(M, z) \int_0^{\lambda^{\rm ob}} d\lambda\, P(\lambda \mid M, z)\, \lambda \cdot 2\pi \int d\theta\, \sin\theta\, f_A(\theta, \lambda^{\rm ob}, z^{\rm ob}, \lambda, z)\, X(M, z, \theta \mid z^{\rm ob}), \tag{21}$$

and X is the per-operator integrand label: $X=1$ for $\mathcal{P}[1]$, $X=b(M,z)\,\xi_{\rm NL}(z,z^{\rm ob},\theta)$ for I_2, and $X=b(M,z)\,\xi_{\rm NL}(z,z^{\rm ob},\theta)\,\sigma(\theta)$ for I_1. $f_X(z)$ is therefore the z-density that the outer integral has to quadrature.

For $\mathcal{P}[1]$, $X=1$ contains no $\xi_{\rm NL}$ and no exclusion; $f_{\mathcal{P}[1]}(z)$ is a smooth single-peaked function of width $\sim\sigma_z(z^{\rm ob})$ peaked at $z=z^{\rm ob}$ (Fig. 5 left panel, dashed blue curve). For I_1, I_2 the halo-exclusion cut carves three regions. Defining $\delta z_{\rm excl}\equiv R_{\rm excl}/(c/H(z^{\rm ob}))$ (at the reference point, $\approx 4.7\times10^{-4}$):

1. *Ring plateau* $|z-z^{\rm ob}|<\delta z_{\rm excl}$: the small-$\theta$ part is excluded; only $\theta>R_{\rm excl}/\chi(z^{\rm ob})$ contributes through a ring of structure at $\Delta\chi>R_{\rm excl}$. Visible as a flat core of $f_{I_2}(z)$ between the dotted lines in Fig. 5 right panel.

2. *Exclusion peaks* at $|z-z^{\rm ob}|\approx\delta z_{\rm excl}$: $\xi_{\rm NL}$ is evaluated at $\Delta\chi\to R_{\rm excl}$, its largest resolved value. $f_{I_2}(z)$ has two sharp peaks at $z=z^{\rm ob}\pm\delta z_{\rm excl}$, a few times the plateau amplitude (Fig. 5 right panel).

3. *Outer decay*: $f(z)\propto\Delta\chi_{\parallel}^{-1.7}$ modulated by w_z, truncating at $|z-z^{\rm ob}|\sim\sigma_z(z^{\rm ob})\approx0.09$ (Fig. 5 left panel, outer wings).

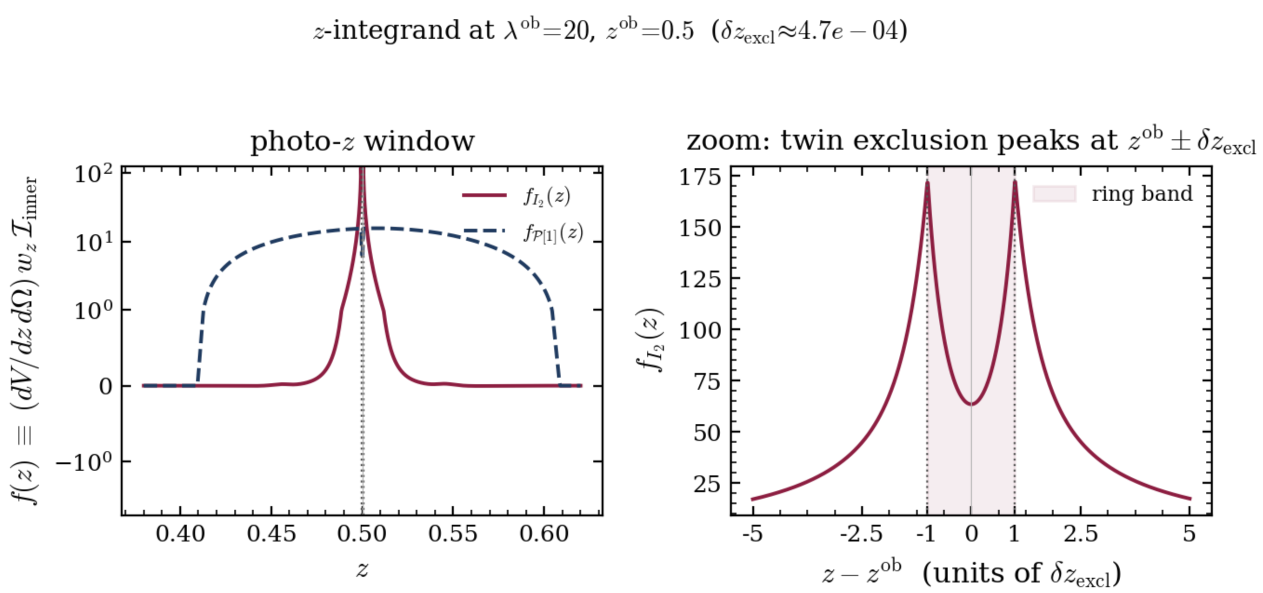

Figure 5: The z-integrand $f_X(z)$ of eq. (20) at $(\lambda^{\rm ob},z^{\rm ob})=(20,0.5)$. *Left:* full photo-$z$ window, symlog y-scale; the red solid curve is $f_{I_2}(z)$, the blue dashed curve is $f_{\mathcal{P}[1]}(z)$. $\mathcal{P}[1]$ (no $\xi_{\rm NL}$) is smooth and single-peaked; I_2 has an additional two-peak exclusion substructure at $z^{\rm ob}\pm\delta z_{\rm excl}$ (dotted vertical lines) that is invisible at this scale but inflated by the symlog axis. *Right:* zoom to $|z-z^{\rm ob}|<5\,\delta z_{\rm excl}$, linear scale. The twin exclusion peaks are resolved; the ring-plateau band between them (red shading) is where $\xi_{\rm NL}$ is excluded and the integrand is flat. A naive uniform z-grid places zero nodes inside this window and under-samples both peaks, which is the hazard the split-by-physics recipe avoids.

Write $I=I_{\rm ring}+I^{\rm fg}_{\rm outer}+I^{\rm bg}_{\rm outer}$ and integrate each region on a grid tailored to its shape. The ring band $(z^{\rm ob}-\delta z_{\rm excl}, z^{\rm ob}+\delta z_{\rm excl})$ uses GL in z (smooth integrand, $N_{\rm ring}=\max(9,\,N_z/4)$). The outer pieces use GL in $u=\ln|\Delta\chi_{\parallel}|$, so $\xi_{\rm NL}\propto e^{-1.7u}$ peaks at the inner endpoint $u=\ln R_{\rm excl}$ where GL naturally clusters nodes. Per side: $N_{\rm outer}=\max(15,\,(N_z-N_{\rm ring})/2)$. Each region's integrand is smooth in its own variable; the sharp features sit at region boundaries where GL rules converge at their design rate.

At $N_z = 80$ this reaches sub-0.01% on $\mathcal{P}[1], I_1, I_2$ vs `scipy.integrate.quad` with matched inner grids (regression test `tests/test_integrals.py::TestSelBiasThetaSplit`). Edge cases ($R_{\rm excl}$ exceeds the photo-z support, or the ring band is wider than it) degrade gracefully via empty-array guards.

`Nz_bias`: a smaller, decoupled budget for `_P_operator`. The $N_z = 80$ budget above is tuned to $\langle \Sigma^{\rm prj} \rangle$'s own needs via `SigmaPrj._build_zM_context` (which calls the same `_z_grid`) and is unchanged. `SelBias._P_operator` now reads a separate `GridConfig.Nz_bias` (default 48) for its own ($\mathcal{P}[1], I_1, I_2$) z-grid. The ring+outer split enforces hard floors ($N_{\rm ring} \geq 9$, $N_{\rm outer} \geq 15$ per side), so any N_z below ~ 40 collapses to the same 39-node grid; 48 sits just above that floor. Against an $N_z = 200, N_\theta = 30$ reference across the 12 DES Y3 ($\lambda^{\rm ob}, z^{\rm ob}$) bins, $N_z = 48$ reaches 0.045% worst-case error (regression test `TestSelBiasNzBias::test_default_Nz_bias_sub_01pct_of_quad`), comfortably inside the 0.1% bar the quad-matched tests enforce, for a measured ~ 1.6–$1.7\times$ reduction in per-call evaluation time (`validations/nz_bias_convergence.py`). An earlier design replaced the ring/outer GL entirely with a polynomial-fit-plus-closed-form scheme (sampling $K \sim 5$–9 nodes and integrating the fit analytically); that was dropped after finding $\xi_{\rm NL}$ goes negative over ~ 20–25% of the outer decay range at higher $z^{\rm ob}$ (BAO-scale separations), which breaks a log-space power-law fit. The existing GL scheme has no such hazard, and — per the floors above — already had slack.

API and glossary

This section is a pointer table from the math above to the code that computes each quantity. The glossary table lists every symbol with its defining equation, a one-line description, and the Python call that returns it.

5 Objects

`Cosmology` (`cosmology.py`). Flat-ΛCDM background. Provides `chi(z)` $= \chi(z)$ comoving distance, `dV_dzdOm(z)` $= dV/dz\, d\Omega$, and `h`, `H0`. Default: Buzzard-v1.1.

`PkGrid` (`pk.py`). CAMB-backed linear $P(k, z)$. One CAMB call per cosmology; re-evaluated in place for each (k, z) request.

`SigmaM` (`sigma_m.py`). $\sigma(M, z)$ top-hat variance via a Simpson rule on `PkGrid`'s k-grid. Used by both `HMF` and `Bias`.

`HMF` (`hmf.py`). Tinker 2008, $\Delta = 200$. `hmf(M, z)` returns $n(M, z)$ in $h^{-1}\, M_\odot^{-1}\, h^{-1}\,\mathrm{Mpc}^{-3}$.

`Bias` (`bias.py`). Tinker 2010 peak-height halo bias. `bias(M, z)` returns $b(M, z)$.

`MOR` (`mor.py`). Mass–observable relation. `mor.pdf(lam, M, z)` returns $P(\lambda \mid M, z)$, the EMG kernel of Costanzi 2019.

`XiNL` (`xi_nl.py`). Nonlinear matter correlation function $\xi_{\mathrm{NL}}(r, z)$ from halofit P_{NL} via an FFTlog Hankel transform. Lazy build (first call pays ~ 2 s, cached thereafter).

`SelBias` (`sel_bias.py`). The bias pipeline (§2.4). Holds references to all of the above. See the glossary below for per-method meanings.

`NFWMiscentered` (`nfw.py`). $\Sigma_{\mathrm{mis}}(R \mid M, z, \theta, z^{\mathrm{ob}})$ from the three offset-NFW lookup tables (Costanzi-notebook convention).

`SigmaPrj` (`sigma_prj.py`). The $\langle\Sigma^{\mathrm{prj}}\rangle$ orchestrator (eq. 16). `SigmaPrj(cosmo, sel_bias, nfw)(R, lob, zob)` returns $\langle\Sigma^{\mathrm{prj}}(R)\rangle$.

`GridConfig` (`config.py`). Grid-size defaults: $N_z = 80$ (read by `SigmaPrj`/ `DeltaSigmaPrj` via `SelBias._z_grid`), $N_z^{\mathrm{bias}} = 48$ (read only by `SelBias._P_operator`, §4), $N_\theta = 10$, $N_M = 24$. N_λ is a separate `SelBias` constructor arg (default 60).

6 Glossary of quantities

Rows group by role: halo ingredients, projection ingredients, the $\mathcal{P}$ operator specialisations, the bias-pipeline outputs, and the profile. “Eq.” links to the defining equation; “Code” gives the Python call.

Symbol	Meaning	Eq.	Code
Halo ingredients			
$n(M,z)$	Tinker-2008 halo mass function ($\Delta = 200$)	(3)	`HMF(sigma_m)(M, z)`
$b(M,z)$	Tinker-2010 peak-height halo bias	(3)	`Bias(sigma_m)(M, z)`
$P(\lambda \mid M,z)$	MOR kernel (Costanzi-2019 EMG)	(8)	`MOR().pdf(lam, M, z)`
$\xi_{\rm NL}(r,z)$	Nonlinear matter 2pt correlation (halofit+FFTlog); zero below $R_{\rm excl}$	(8)	`XiNL(cosmo)(r, z)`
Projection kernel and $\mathcal{P}[X]$ operator			
$\rho^{\rm prj}$	Per-halo projection weight $w_z\, f_A\, \lambda$ ($\lambda < \lambda^{\rm ob}$)	(5)	internal to `SelBias._P_operator`
$\mathcal{P}[X]$	LoS average of X weighted by $\rho^{\rm prj}$ and halo distribution	(8)	`SelBias._P_operator(lob, zob)`
$\mathcal{P}[1] \equiv \langle \Delta^{\rm prj}_{\rm bkg} \rangle$	Mean background projection (no $\xi_{\rm NL}$, smooth in z)	(8)	`bias_precompute(lob,zob)['P1']`
$I_2 \equiv \mathcal{P}[\,b\,\xi_{\rm NL}\,]$	Halo-bias-weighted $\xi_{\rm NL}$; drives $\Delta^{\rm prj}_{\rm RND}$	(8)	`bias_precompute(lob,zob)['I2']`
$I_1 \equiv \mathcal{P}[\,b\,\xi_{\rm NL}\,\sigma(\theta)\,]$	I_2 weighted by the sigmoid; enters Step 5	(8)	`bias_precompute(lob,zob)['I1']`
Bias-pipeline outputs			
$b_{\rm halo}(\lambda^{\rm ob}, z^{\rm ob})$	Halo-bias aggregate at fixed $\lambda^{\rm ob}$ (paper's $b_{\rm eff}$)	(3)	`SelBias.b_eff(lob, zob)`
$\Delta^{\rm prj}_{\rm RND}$	Projection contamination of a random aperture	(10)	`bias_precompute(lob,zob)['Delta_RND']`
$b^{\rm large}_{\rm sel}$	Large-θ plateau (empirical closure)	(11)	`bias_from_precomp(pre,ltr)['b_infty']`
$b^{\rm small}_{\rm sel}$	Small-θ plateau (Step 5 inversion)	(12)	`bias_from_precomp(pre,ltr)['b_zero']`
$b_{\rm sel}(\lambda^{\rm ob}, \lambda^{\rm tr}, z^{\rm ob}, \theta)$	Scale-dependent bias via sigmoid ansatz	(4)	`SelBias.b_lob_theta(th,ltr,zob,lob,precom`
$\bar{b}_{\rm small}, \bar{b}_{\rm large}$	$\lambda^{\rm tr}$-averaged plateaus (cached per $(\lambda^{\rm ob}, z^{\rm ob})$)	(14)	`SelBias._marginalised_plateaus(lob,zob,pre`
$b_{\rm sel}(\lambda^{\rm ob}, z^{\rm ob}, \theta)$	$\lambda^{\rm tr}$-marginalised bias; input to $\langle \Sigma^{\rm prj} \rangle$	(15)	`SelBias.b_sel_marginalised(th,lob,zob,pre`
Two-halo profile			
$\Sigma_{\rm mis}(R \mid M, z, \theta, z^{\rm ob})$	Azimuth-averaged miscentered NFW (offset-NFW lookup)	(17)	`NFWMiscentered(cosmo).sigma_grid(R,Rtheta`
$\Delta\Sigma_{\rm mis}$	Miscentered NFW radial excess $\bar{\Sigma}_{\rm mis}(<R) - \Sigma_{\rm mis}(R)$ (companion lookup table)	—	`NFWMiscentered.delta_sigma_grid`
$\langle \Sigma^{\rm prj}(R \mid \lambda^{\rm ob}, z^{\rm ob}) \rangle$	Two-halo projected surface density (main output)	(16)	`SigmaPrj(cosmo,sel_bias,nfw)(R,lob,zob)`
$\langle \Delta\Sigma^{\rm prj}(R \mid \lambda^{\rm ob}, z^{\rm ob}) \rangle$	Two-halo lensing excess (see `docs/delta_sigma_prj_derivation.tex`)	—	`DeltaSigmaPrj(cosmo,sel_bias,nfw)(R,lob,z`

7 Validation

Unit tests. `tests/test_integrals.py`: `TestSelBiasThetaSplit` (sub-0.1% vs quad at $N_\theta = 10$), `TestLambdaAxisConverged` ($n^{\rm tr}_\lambda = 30$ matches 100 to $< 10^{-4}$), `TestSigmaPrjTheta` (split-at-θ_R +

monotonicity). `test_smoke.py`: end-to-end smoke test.

Validation scripts. `validations/`: `theta_convergence.py` sweeps N_θ; `quad_matched_inner.py` compares the `_P_operator` output to `scipy.integrate.quad` with matched inner GL grids; `quad_validate.py` is the end-to-end quad cross-check.

Defaults. $N_z = 80$, $N_\theta = 10$, $N_M = 24$, $N_\lambda = 60$. No retuning is needed across the Y3 range ($\lambda^{\rm ob} \in [20, 500]$, $z^{\rm ob} \in [0.1, 0.8]$): only $R_{\rm excl}$ and $c/H(z^{\rm ob})$ shift, and both are tracked analytically by the grid builder.

References

Costanzi, M., Wu, H.-Y., Esteves, J. H., Grandis, S., To, C., & Aguena, M. 2026, Phys. Rev. D (accepted), arXiv:2604.05833.